

EDU AI – STUDENT PERFORMANCE & DROPOUT PREDICTION SYSTEM

PHASE I REPORT

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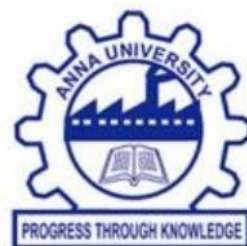
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“This project addresses the following Sustainable Development Goal(s): SDG 4: Quality Education; SDG 9: Industry, Innovation and Infrastructure”

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ABSTRACT

EDU AI is a Student Performance & Dropout Prediction System designed to help educational institutions identify students who are at risk of poor academic performance or dropout at an early stage. The system analyzes various student-related factors, including attendance, assignment and quiz scores, examination marks, study hours, previous GPA, demographics, extracurricular participation and historical dropout records. Using artificial intelligence and machine learning techniques, it predicts student academic performance, identifies dropout risk and generates an individual risk score for every student. An analytics dashboard provides educators with insights into student progress and highlights students who may require timely academic intervention.

The proposed system integrates multiple machine learning algorithms, namely Logistic Regression, Random Forest, Naïve Bayes and K-Means Clustering, to analyze patterns in historical and current student data and generate predictions. The risk assessment mechanism categorizes students into Low, Medium and High risk levels, enabling faculty members to prioritize students who require greater attention. The system also provides data visualization and reporting features that make academic trends and risk factors easier to understand. By reducing dependence on manual monitoring and enabling data-driven decision-making, EDU AI aims to support proactive intervention, improve student retention and enhance overall academic success. The project focuses on performance prediction, dropout prediction, risk scoring, analytics and reporting, while automatic grading, course recommendations, admission management and examination administration are outside its scope.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
DBMS	Database Management System
DFD	Data Flow Diagram
EDM	Educational Data Mining
ERP	Enterprise Resource Planning
GPA	Grade Point Average
LMS	Learning Management System
LR	Logistic Regression
ML	Machine Learning
NB	Naïve Bayes
RF	Random Forest
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Education is a long-term process in which the academic outcome of a student depends on many interacting factors such as class attendance, regular completion of assignments, performance in quizzes and examinations, time devoted to self-study, previous academic record, personal background and participation in co-curricular activities. When these factors begin to deteriorate, the student is often already on a path towards poor academic performance or, in serious cases, towards leaving the programme altogether. Dropout is costly for the student, whose career opportunities are affected, and for the institution, whose retention and academic quality are measured by it.

In most institutions, student performance is monitored through attendance registers, internal assessment records and end-of-semester results. These records are valuable, but they are usually examined only after a problem has become visible, for example after a failed examination or a prolonged absence. A struggling student can therefore remain unnoticed for weeks or months, during which timely guidance would have been most effective. Early identification of such students, followed by prompt academic support and counselling, is widely regarded as one of the most effective ways of improving student retention.

Educational Data Analytics applies data analysis and machine learning techniques to the data generated in academic environments. Machine learning models can learn from historical student records to detect patterns that separate students who succeeded from those who struggled or dropped out, and can then apply those patterns to current students. Such models are able to consider many factors simultaneously, which is difficult to do by manual inspection.

EDU AI builds on this idea. It records student academic and behavioural data, prepares it for analysis, applies machine learning models to predict academic performance and dropout risk, converts the result into an individual risk score and groups students into Low, Medium and High risk levels. The results are presented to faculty and academic administrators

through an analytics dashboard and reports, so that attention can be directed to the students who need it most and decisions can be based on data rather than on intuition alone.

1.2 PROBLEM STATEMENT

Educational institutions collect large amounts of student data, but this data is frequently scattered across attendance sheets, assessment records and administrative files. Faculty members are expected to monitor the progress of many students at the same time, and it is difficult for them to notice the combined effect of falling attendance, declining quiz scores, reduced study effort and a weak previous academic record on any single student. As a result, monitoring is largely manual, reactive and dependent on the experience of individual faculty members.

Students who are at risk of poor performance or dropout often show early warning signs that are individually small and therefore easy to overlook. By the time the risk becomes obvious, for example through failure in an examination or prolonged absence, the opportunity for effective and low-cost intervention may already have been lost. Existing reports are mostly descriptive: they show what has happened but do not estimate what is likely to happen, and they do not rank students by the level of risk they face.

There is therefore a need for an AI-driven system that analyzes academic and behavioural student data together, predicts academic performance and dropout risk, produces an understandable individual risk score and risk level, and presents the results in a form that faculty and administrators can act upon. Such a system would reduce dependence on manual monitoring, support early intervention and contribute to improved student retention and academic success.

1.3 EXISTING SYSTEM

In most institutions, student monitoring is carried out through manually maintained attendance registers, spreadsheets of internal marks and periodic reviews by class advisors or mentors. These methods record what has already happened and depend heavily on the time and attention that individual faculty members can devote to each student. Information about a single student is held in several places, and combining it to form an overall picture is slow and error-prone.

Where institutions use software for academic records, the typical output consists of attendance percentages, mark lists and summary reports. These tools are useful for record keeping and reporting, but they generally do not analyze the relationship between multiple factors, do not learn from historical dropout cases and do not estimate the likelihood that a given student will perform poorly or drop out. Identification of students who need help therefore relies on fixed cut-off rules, such as a minimum attendance requirement, or on informal observation.

Because of these characteristics, existing practice tends to identify academically weak students late, treats all students according to the same fixed thresholds, offers limited centralized analytics and provides little support for prioritizing interventions. It also provides no consistent, quantified measure of how serious a student's situation is, which makes it difficult to compare students or to monitor whether an intervention has been effective.

1.4 PROPOSED SYSTEM

EDU AI is a Student Performance & Dropout Prediction System that analyzes student data using machine learning to identify students at risk of poor academic performance or dropout at an early stage. It is a connected analytics system that combines student profiles, academic records, behavioural indicators, historical dropout records, prediction models, risk scoring and a faculty-oriented dashboard.

The system begins with student data collection and profile management. Attendance, assignment scores, quiz scores, examination marks, study hours, previous GPA, demographic information, extracurricular participation and historical dropout records are validated and stored against each student. The stored data is then cleaned, encoded and scaled during preprocessing, and informative features are selected or derived during feature engineering so that the data is suitable for machine learning.

The prepared data is given to the machine learning engine, which uses four models: Logistic Regression for performance prediction, Random Forest for dropout risk classification, Naïve Bayes for behavioural risk assessment and K-Means Clustering for student segmentation. The engine produces two main predictions for every student: an academic performance prediction and a dropout risk prediction. These are expressed as an individual risk score, and the score is mapped to one of three risk levels: Low Risk, Medium

Risk or High Risk. Each result is accompanied by the main factors that contributed to the risk, so that faculty can understand why a student has been flagged.

Finally, an analytics dashboard presents student performance trends, attendance analytics, risk distribution, prediction results and lists of at-risk students, and a reporting function summarizes these findings for faculty and academic administrators. This enables early, targeted and data-driven intervention instead of reaction after failure has occurred. The basic architecture of EDU AI is shown in Figure 1.1.

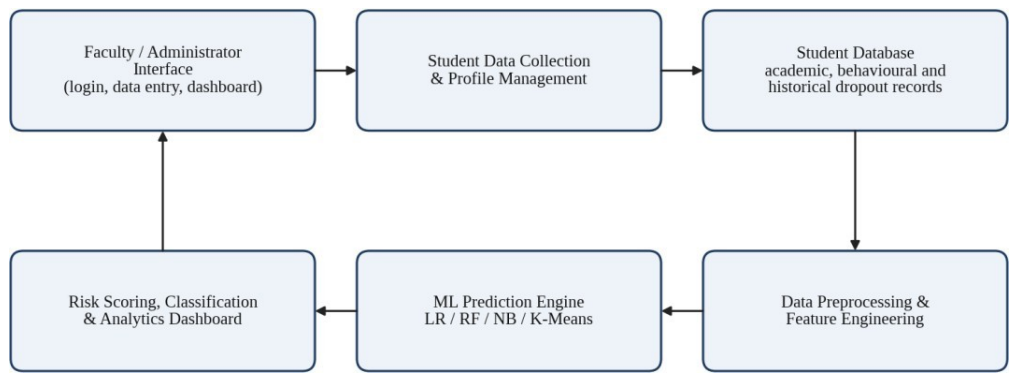


Figure 1.1 EDU AI basic architecture

1.5 OBJECTIVES

The main objective of EDU AI is to identify students who are at risk of poor academic performance or dropout at an early stage, using student-related academic and behavioural data. The system is intended to provide faculty and academic administrators with predictions, individual risk scores and analytics that support timely and well-directed academic intervention. Each outcome must be traceable to the student data supplied and to the output of the machine learning models, as listed in Table 1.1.

Table 1.1 Project objectives

NO.	OBJECTIVE
1	Collect and manage student profiles and academic records, including attendance, assignment scores, quiz scores, examination marks, study hours, previous GPA, demographics, extracurricular participation and historical dropout records.
2	Preprocess student data and engineer features suitable for machine learning.
3	Predict student academic performance using machine learning models.
4	Predict the dropout risk of each student using machine learning models.
5	Generate an individual risk score for each student and classify students into Low, Medium and High risk levels.
6	Provide an analytics dashboard with data visualizations that present performance trends, risk distribution and prediction results.
7	Generate reports and identify at-risk students early so that faculty can plan timely intervention.

1.6 SCOPE

The scope of EDU AI covers student data collection and profile management, data preprocessing and feature engineering, student performance prediction, dropout prediction, individual risk scoring, risk-level classification, student analytics, data visualization, academic reporting and support for early intervention. The primary users are faculty members and academic administrators of an educational institution, and the system analyzes student data and provides predictions and analytics to these authorized users.

The system focuses on prediction, risk assessment and analytics. Automatic grading, course recommendations, admission management and examination administration are deliberately excluded from the scope of this project. The predictions and risk levels are intended as decision support for faculty and do not replace professional academic judgement.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Educational Data Mining. Educational Data Mining (EDM) is the application of data mining and statistical techniques to data produced in educational settings, such as attendance logs, assessment records and learning-management activity. Its aim is to discover patterns that explain how students learn and why some students succeed while others struggle. Typical EDM tasks include classification of students, clustering of similar learners and prediction of academic outcomes. A common limitation noted in this field is that many studies stop at analysis and do not deliver the findings to teachers in a form they can use directly. EDU AI addresses this by pairing prediction with a faculty dashboard and risk levels. [1], [2]

Student Performance Prediction. Studies on student performance prediction use attributes such as previous grades, attendance, assessment scores and study behaviour to estimate a student's future marks, grade or pass/fail status. Both classification formulations (for example, pass or fail) and regression formulations (for example, a predicted score) are common. These studies generally find that earlier academic achievement and engagement indicators carry strong predictive information. Limitations often reported include dependence on a single course or institution and limited attention to how predictions are presented to teachers. EDU AI combines several academic and behavioural attributes and presents the prediction together with its contributing factors. [3], [4]

Student Dropout Prediction. Dropout prediction is usually treated as a binary classification problem in which the model estimates whether a student will leave a programme before completion. Researchers commonly combine academic indicators with demographic and behavioural attributes, and many note that dropout cases form a small proportion of the data, which creates class imbalance. Such imbalance can make a model appear accurate while still missing many students who actually drop out. EDU AI therefore treats dropout prediction as a distinct task from performance prediction and emphasizes evaluation measures beyond accuracy. [2], [3], [5]

Machine Learning in Education. Logistic Regression, Random Forest, Naïve Bayes and K-Means Clustering are among the techniques frequently applied to educational data. Logistic Regression and Naïve Bayes are valued for simplicity, efficiency on small datasets and interpretability, Random Forest is an ensemble method that often captures non-linear relationships and feature interactions more effectively, and K-Means Clustering is an unsupervised method used to group similar students. Studies generally compare several algorithms on the same data because no single algorithm is best in every setting. EDU AI follows this practice by using four algorithms for different roles and comparing their results. [1], [8]

Early Warning Systems. Early warning systems in education aim to flag students who show indicators of academic difficulty early enough for support to be effective. They are typically built on a small set of indicators such as attendance and assessment performance, and may use simple threshold rules or predictive models. Rule-based systems are easy to understand but cannot capture the combined effect of many factors, while predictive systems can but must be made understandable to staff to be trusted. EDU AI aims to combine the predictive strength of machine learning with clear risk levels and visible contributing factors. [6], [7]

Academic Risk Prediction and Risk Scoring. Academic risk prediction goes beyond a yes/no outcome by expressing the degree of risk, either as a probability or as a score or category. Risk levels allow staff with limited time to prioritize the students who need the most attention. A limitation is that the meaning of a score or category must be defined carefully and explained to users, otherwise it may be misinterpreted. EDU AI generates an individual risk score and classifies it into Low, Medium and High risk levels with an explanation of contributing factors. [4], [6]

Student Analytics and Dashboards. Learning analytics and student analytics research highlights the value of visual dashboards that summarize attendance, assessment trends and risk for teachers and administrators. Effective dashboards are those that lead to a decision or action rather than simply displaying data. EDU AI includes a dashboard that presents performance trends, attendance analytics, risk distribution and at-risk student lists, and a reporting function for academic review. [7]

Predictive Analytics in Education. Predictive analytics in education brings together prediction models, institutional data and decision making. Literature in this area also raises important concerns, including data privacy, fairness across student groups, the risk of labelling students and the need for transparent models. These concerns suggest that predictions should be used as decision support for educators rather than as automatic decisions. EDU AI is designed as decision support for authorized faculty and administrators. [5]

2.2 COMPARISON AND DISCUSSION

The areas reviewed above provide a sound but fragmented foundation for EDU AI. Table 2.1 summarizes the general focus of each area and the way in which EDU AI builds on it.

Table 2.1 Comparison of literature themes with EDU AI

THEME	TYPICAL FOCUS	CONTRIBUTION OF EDU AI
Educational Data Mining	Discovering patterns in educational data	Applies EDM to produce actionable predictions and risk levels
Performance prediction	Estimating future marks, grades or pass/fail	Predicts performance from academic and behavioural factors together
Dropout prediction	Binary prediction of leaving a programme	Treats dropout as a separate prediction task with a risk score
Machine learning in education	Comparing algorithms on educational data	Supports Logistic Regression, Random Forest, Naïve Bayes and K-Means Clustering
Early warning systems	Flagging students early using indicators	Adds predictive models and graded risk levels
Student analytics	Dashboards for teachers and administrators	Provides risk-oriented dashboards and reports
Predictive analytics	Decision support, ethics and fairness	Positions predictions as decision support for authorized users

Existing work tends to concentrate on one aspect, such as a single prediction task, a comparison of algorithms or a dashboard. EDU AI brings these aspects into a single, faculty-

centred system that links student data, preprocessing, machine learning prediction of performance and dropout, individual risk scoring, risk classification, analytics and reporting.

2.3 CONCLUSION

The survey of the research themes confirms the value of machine learning and data analytics for identifying students who need support, and shows the need for a system that combines performance prediction, dropout prediction, risk scoring and analytics in one place. It also highlights the importance of interpretable results, careful handling of class imbalance, responsible use of student data and presentation of predictions in a form that educators can act upon. EDU AI addresses this need by providing an integrated platform for early identification of at-risk students and for data-driven academic decision making.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM LIMITATIONS

EDU AI was created to overcome the shortcomings of conventional methods of monitoring student progress. The main limitations of the existing approach are described below.

- **Manual student monitoring.** Attendance and marks are tracked in registers and spreadsheets, so monitoring depends on the time faculty can spare and grows harder as class sizes increase.
- **Delayed identification of academically weak students.** Problems are often noticed only after an examination result or a long absence, when early support is no longer possible.
- **Difficulty analyzing multiple student factors.** Attendance, assignments, quizzes, examinations, study hours, previous GPA and background interact, and it is hard to judge their combined effect manually.
- **Lack of centralized analytics.** Student information is spread across different records, so no single view of progress and risk is available to faculty or administrators.
- **Lack of predictive risk assessment.** Reports are descriptive and use fixed cut-offs; they do not estimate how likely a student is to perform poorly or drop out.
- **Difficulty identifying potential dropout cases early.** Historical dropout records are rarely used to learn which patterns preceded earlier dropouts.
- **Limited data-driven intervention.** Without a ranked list of at-risk students, support is distributed unevenly and the results of interventions are hard to follow.

EDU AI addresses these limitations through a connected student database, a data preprocessing and feature engineering pipeline, machine learning models for performance and dropout prediction, individual risk scores with Low, Medium and High risk levels, and an analytics dashboard with reporting.

3.2 PROPOSED SYSTEM ARCHITECTURE

The proposed EDU AI architecture is shown in Figure 3.1. It follows a pipeline from student data to faculty decision support. Student data covering attendance, assignment scores, quiz scores, examination marks, study hours, previous GPA, demographics, extracurricular participation and historical dropout records is collected through the student data input and stored in the student database.

The data preprocessing stage cleans the data, handles missing values, encodes categorical variables and scales numerical values. The feature engineering stage then selects relevant features and derives additional academic and behavioural features. The prepared feature set is passed to the ML prediction engine, which uses Logistic Regression for performance prediction, Random Forest for dropout risk classification, Naïve Bayes for behavioural risk assessment and K-Means Clustering for student segmentation.

The engine runs two prediction tasks: performance prediction and dropout risk prediction. Their outputs are combined by the risk score generation stage into an individual risk score for each student. The risk classification stage maps the score to Low Risk, Medium Risk or High Risk. Finally, the analytics dashboard presents trends, risk distribution, prediction results and at-risk students to the faculty member or academic administrator, who decides on the appropriate academic intervention.

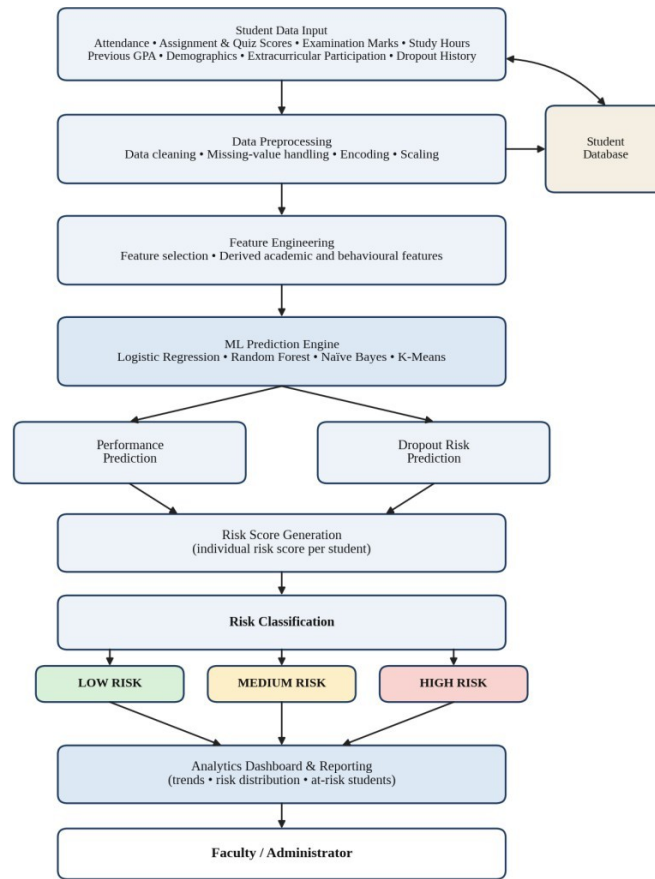


Figure 3.1 EDU AI system architecture

3.3 SYSTEM REQUIREMENTS

The system shall allow authorized faculty and administrators to enter or upload student data and to maintain student profiles. It shall validate the data before storing it, including checks for missing, out-of-range or inconsistent values in attendance, scores, marks, study hours and GPA. It shall preprocess the data and derive features, predict student performance, predict dropout risk, generate an individual risk score and classify each student as Low, Medium or High risk. It shall present the results through an analytics dashboard and reports, and shall make predictions and analytics available only to authorized users.

3.3.1 SOFTWARE REQUIREMENTS

The software requirements are summarized in Table 3.1. Specific tools and versions will be entered once the implementation environment is finalized.

Table 3.1 Software requirements

CATEGORY	REQUIREMENT
Operating System	Windows or Linux [TO BE PROVIDED]
Programming Language	[TO BE PROVIDED] – for data processing, model training and prediction
Machine Learning Libraries	[TO BE PROVIDED] – for Logistic Regression, Random Forest, Naïve Bayes and K-Means Clustering
Data Handling Libraries	[TO BE PROVIDED] – for cleaning, transformation and analysis of student data
Visualization	[TO BE PROVIDED] – for dashboard charts and report visualizations
Database	[TO BE PROVIDED] – a DBMS for student, prediction and risk records
User Interface	[TO BE PROVIDED] – interface for faculty and administrators
Web Browser	Latest version of a modern web browser, if the dashboard is web-based [TO BE PROVIDED]

3.3.2 HARDWARE REQUIREMENTS

The system is intended to run on standard personal computer hardware during development. The indicative requirements are given in Table 3.2 and may be revised when the final deployment environment is decided. Model training on larger student datasets will benefit from additional memory and processing capacity.

Table 3.2 Hardware requirements

CATEGORY	MINIMUM REQUIREMENT	RECOMMENDED REQUIREMENT
Processor	Dual-core CPU	Quad-core CPU or better
Memory	4 GB RAM	8 GB RAM or more
Storage	10 GB free space	SSD with 20 GB or more free space
Network	Stable internet connection	High-speed internet connection
Display	1366 × 768 resolution	Full HD display
Client Device	Desktop or laptop	Desktop or laptop with a modern browser

3.3.3 DATABASE DESIGN

The database design is centred on the student entity. Each student has one profile and many academic records. An academic record stores attendance, assignment scores, quiz scores, examination marks and GPA, while behavioural data stores study hours and extracurricular participation. Historical dropout records are stored for model training. For each student the system also stores prediction results, risk scores with risk levels and contributing factors, and analytics summaries used by the dashboard and reports.

Relationships between entities are maintained through the student identifier so that every record can be traced to its student, and so that prediction and risk results can be retained over time for trend analysis. Access to student records is restricted to authorized faculty and administrators. The entities and key data are listed in Table 3.3. The choice of DBMS is [TO BE PROVIDED].

Table 3.3 Student data dictionary

ENTITY	KEY DATA
Student	Student ID, name, age, gender, department
Academic Record	Attendance, assignments, quizzes, exams, GPA
Behavioural Data	Study hours, extracurricular participation
Prediction	Performance prediction, dropout prediction
Risk Score	Score, risk level, contributing factors
Analytics	Trends, summaries, reports

3.4 SYSTEM DESIGN

The system design follows the principle of a single source of truth for student data. Data entered or uploaded by an authorized user is validated and stored in the student database. All later stages, namely preprocessing, prediction, risk scoring and analytics, read from this stored data, and their results are stored again so that the dashboard and reports always present consistent information.

3.4.1 DATA FLOW DIAGRAM

The data flow of EDU AI is shown in Figure 3.2. The faculty member or administrator supplies student data through the student data entry process. The student data validation process checks the data; invalid data is returned to the user with a validation error, and valid data is stored in the student database (D1). The data preprocessing and feature engineering process reads student records, produces an engineered feature set and stores it in the feature store (D2).

The ML prediction module reads the feature set to predict student performance, and the dropout risk prediction process uses the same features together with the predicted performance to estimate dropout risk. The risk score generation and classification process combines these outputs into a risk score and a Low, Medium or High risk level, which are stored in the prediction and risk records (D3). The analytics dashboard reads the stored results and presents dashboards, reports and the at-risk student list to the faculty member or administrator.

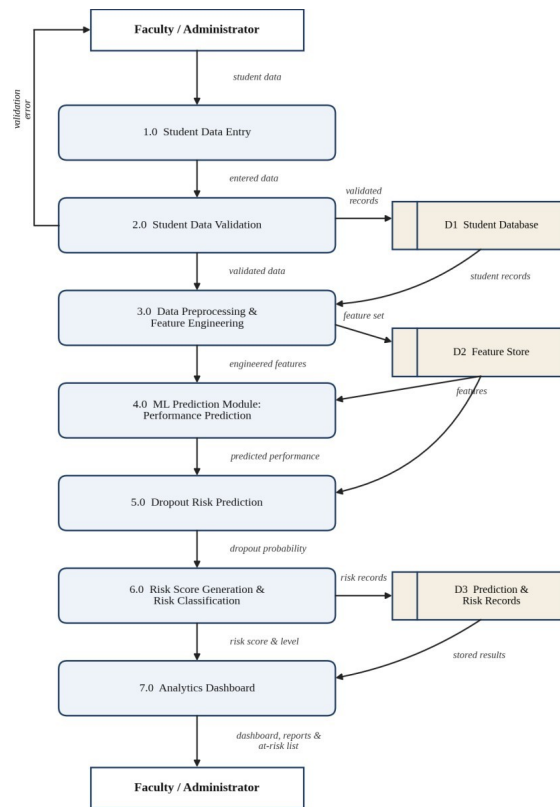


Figure 3.2 EDU AI data flow diagram

3.4.2 ACTIVITY DIAGRAM

The activity begins when a faculty member or administrator logs in to the system and enters or uploads student data. The system validates the data. If the data is not valid, a validation error is shown and the user enters the data again. If the data is valid, the system preprocesses the data, extracts features and runs the machine learning models.

The models predict student performance and dropout risk, after which the system generates a risk score and classifies the risk as Low, Medium or High. The activity ends when the dashboard is displayed with the results. The flow is shown in Figure 3.3.

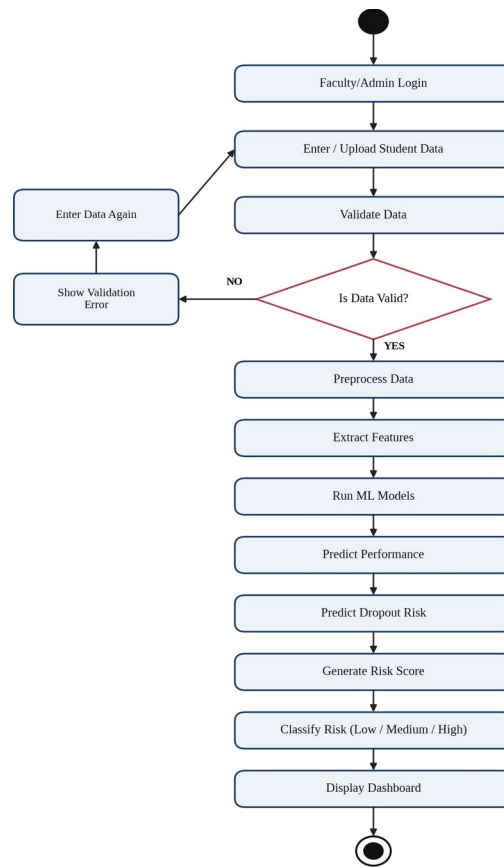


Figure 3.3 EDU AI activity diagram

CHAPTER 4

MODULE DESCRIPTION

4.1 STUDENT PROFILE & DATA MANAGEMENT

This module manages the student information and academic data required by all other modules. Authorized faculty and administrators can enter or upload data and maintain student profiles. Each student is identified by a unique identifier, and all academic, behavioural and historical records are linked to it, so that the information about a student is available in one place.

The module stores student information such as identifier, name, age, gender and department, and the academic records of the student, including attendance, assignment scores, quiz scores, examination marks and previous GPA. It also stores behavioural data such as weekly study hours and extracurricular participation, demographic information, and historical records, including historical dropout records used for training the prediction models. The data stored by this module is summarized in Table 4.1.

Table 4.1 Student data features

CATEGORY	REPRESENTATIVE FEATURES	PURPOSE
Student information	Student ID, name, age, gender, department	Identifies the student and links all records
Attendance	Attendance records or attendance percentage	Indicates regularity and engagement
Assessment data	Assignment scores, quiz scores, examination marks	Indicates academic achievement
Previous academic record	Previous GPA	Indicates past academic standing
Study behaviour	Study hours	Indicates academic effort
Demographic information	Demographic attributes of the student	Provides background context
Extracurricular participation	Participation in extracurricular activities	Indicates wider engagement
Historical records	Historical dropout records	Supplies outcomes for model training

Before data is stored it is validated. Attendance and scores must lie within valid ranges, study hours must be non-negative, identifiers must be unique and mandatory fields must be present. Records that fail validation are returned to the user with a message so that they can be corrected. The module separates stored data from data derived by the system, such as predictions and risk scores, so that raw student data remains unchanged by later processing.

Student records are accessed only by authorized users. Because all later modules read the same validated records, the dashboard, predictions and reports remain consistent with each other.

4.2 DATA PREPROCESSING & FEATURE ENGINEERING

Raw student data is rarely ready for machine learning. It may contain missing values, inconsistent formats, categorical attributes and values measured on different scales. This module converts the stored data into a clean, numerical and informative feature set. The main operations are listed in Table 4.2.

Table 4.2 Preprocessing and feature engineering operations

OPERATION	DESCRIPTION
Data cleaning	Remove duplicate records, correct inconsistent formats and handle invalid or outlying values.
Missing-value handling	Treat missing attendance, scores or other fields, for example by imputation or by flagging the record, depending on the field.
Encoding categorical variables	Convert attributes such as gender, department and extracurricular participation into numerical form.
Feature selection	Retain features that contribute to prediction and remove redundant or uninformative ones.
Data transformation	Derive features such as average assignment score, average quiz score or change in GPA from the stored data.
Normalization / scaling	Bring numerical features to comparable scales where required, for example for Logistic Regression.
Data preparation	Arrange the final feature set and split it into training and testing portions for model development.

Missing values are handled according to the nature of the field, so that incomplete records do not have to be discarded unnecessarily. Categorical variables are encoded because machine learning algorithms operate on numerical input. Scaling is applied where the algorithm is sensitive to feature magnitude, such as Logistic Regression, while tree-based models are generally less sensitive to scale.

Feature engineering combines basic fields into measures that describe student behaviour more directly. Examples include an overall assessment average, the trend of quiz or assignment scores across the term, and the relationship between study hours and results. Which derived features are finally used is decided during model development. Careful selection of features reduces noise, shortens training and makes the resulting model easier to interpret.

4.3 STUDENT PERFORMANCE PREDICTION

The Student Performance Prediction module uses the engineered feature set to predict how a student is likely to perform academically. A Logistic Regression model learns the

relationship between student features, such as attendance, assignment, quiz and examination results, study hours and previous GPA, and recorded academic outcomes from historical data. After training, the model is applied to current students to produce a performance score, expressed as a percentage, for each of them.

EDU AI uses four algorithms, each assigned a specific role. Their relevance to student performance and dropout prediction is summarized in Table 4.3.

Table 4.3 Relevance of machine learning models

MODEL	DESCRIPTION	ROLE IN EDU AI
Logistic Regression	Predicts a score using a weighted linear combination of features passed through a sigmoid activation. Each coefficient directly represents the contribution of a feature.	Performance prediction
Random Forest	Ensemble of 10 decision trees trained on random feature subsets (bagging). Each tree votes on the dropout risk class and the majority wins.	Dropout classification
Naïve Bayes	Applies Bayes' theorem with Gaussian likelihoods for continuous features and class priors from the dataset distribution. Efficient and effective with small datasets.	Behavioural risk assessment
K-Means Clustering	Unsupervised algorithm that partitions the students into $K = 4$ clusters by minimizing the within-cluster sum of squared Euclidean distances. Pre-converged centroids are used for reproducibility.	Student segmentation

Logistic Regression computes the performance score by passing a weighted linear combination of the student features through the sigmoid function and scaling the result to a percentage:

$$z = \sum \beta_i x_i + \beta_0, \sigma(z) = 1 / (1 + e^{-z}), \text{ Score \%} = \sigma(z) \times 100$$

where β_0 is the intercept and β_i are coefficients learned from the data. Because each coefficient shows how strongly a feature affects the score, the model is useful for explaining results to faculty. Random Forest classifies the dropout risk class of a student by majority vote of its trees:

$$RF(x) = \operatorname{argmax}_c \sum_k I(T_k(x) = c)$$

where T_k is the k -th tree and I is an indicator function that equals 1 when the tree votes for class c . Naïve Bayes assigns a student to the class with the highest posterior probability, assuming Gaussian likelihoods for continuous features:

$$P(c | X) \propto P(c) \cdot \prod_i P(x_i | c)$$

K-Means Clustering is unsupervised and groups similar students by assigning each student to the nearest cluster centroid:

$$cluster(x) = \underset{k}{\operatorname{argmin}} \|x - \mu_k\|^2$$

The four models are trained on the student data and their results are compared, as presented in Chapter 5. Figure 4.1 shows the training and prediction workflow of EDU AI.

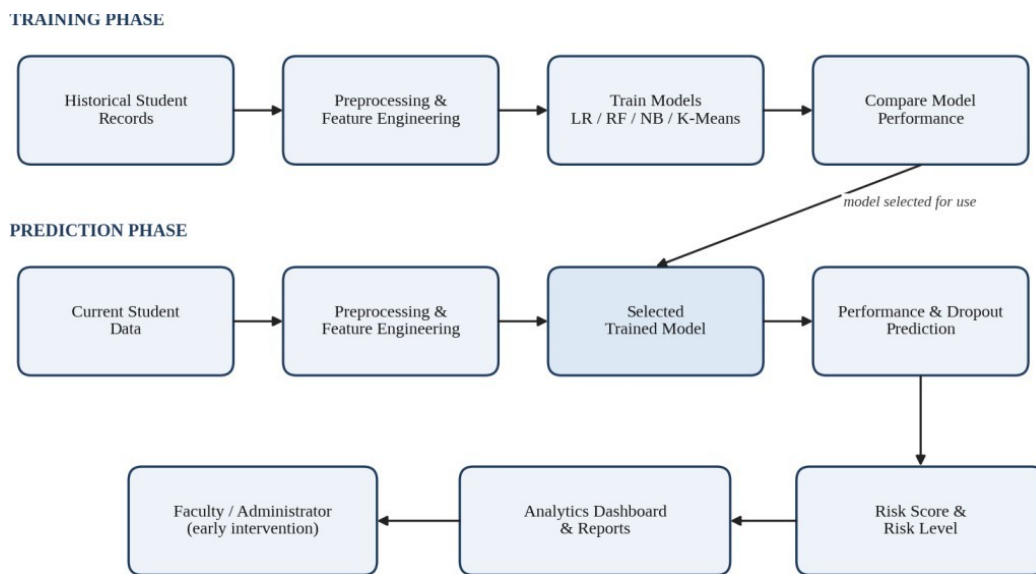


Figure 4.1 EDU AI prediction workflow

4.4 DROPOUT RISK PREDICTION & RISK SCORING

The Dropout Risk Prediction module estimates the likelihood that a student will leave the programme. It is a separate task from performance prediction because a student with modest marks may continue successfully, while a student with other warning signs may still be at risk. The module uses a Random Forest classifier, an ensemble of 10 decision trees in which each tree votes on the dropout risk class and the majority decides. It learns from historical dropout records together with academic and behavioural data and applies the learned patterns to current students. A Naïve Bayes model supports the assessment of behavioural risk, and K-Means Clustering groups students into four clusters for student segmentation.

Because dropout cases are usually fewer than non-dropout cases, the module takes care in training and evaluating the models. Measures such as precision, recall and F1-score, and not accuracy alone, are used to judge how well at-risk students are identified.

Risk Score Generation

The risk score is an individual value that summarizes the performance prediction and the dropout risk estimate for a student. Conceptually it can be expressed as a function of the model outputs and the main student indicators:

$$\text{Risk Score} = f(\text{predicted performance, dropout risk estimate, key indicators})$$

The risk score is expressed as a percentage from 0 to 100. The exact formula and the weight given to each component are [TO BE PROVIDED]. The components that make up the risk score are described in Table 4.4.

Table 4.4 Student risk score components

COMPONENT	INPUTS	ROLE IN THE RISK SCORE
Academic performance	Predicted performance based on assignments, quizzes, examinations and previous GPA	Indicates risk from weak academic results
Attendance	Attendance records	Indicates risk from low engagement
Study behaviour	Study hours, extracurricular participation	Indicates risk from low effort or disengagement
Dropout probability	Output of the dropout prediction models using historical dropout patterns	Indicates likelihood of leaving the programme
Background factors	Demographic information	Provides context for the prediction

Risk-Level Classification

The risk score is mapped to one of three levels so that faculty can quickly see which students need attention. A score below 50% is classified as Low Risk (shown on the dashboard as Stable), a score from 50% to 80% as Medium Risk (Moderate) and a score above 80% as High Risk (Critical). The levels and their interpretation are given in Table 4.5.

Table 4.5 Student risk level classification

RISK LEVEL	INTERPRETATION	SUGGESTED FACULTY ATTENTION
LOW RISK	Risk score below 50% (Stable). The student shows no major warning signs and is likely to continue performing satisfactorily.	Routine monitoring.
MEDIUM RISK	Risk score of 50% to 80% (Moderate). The student shows some warning signs that may worsen without attention.	Closer monitoring and early academic support.
HIGH RISK	Risk score above 80% (Critical). The student shows several strong warning signs and is at a high likelihood of poor performance or dropout.	Priority attention and timely intervention by faculty.

Factors Contributing to Risk

For every flagged student the module identifies the factors that contributed most to the risk, for example low attendance, declining quiz scores, reduced study hours or a low previous GPA. This is obtained from model outputs such as feature importance or model coefficients. Showing the reasons makes the risk level understandable and helps faculty choose a suitable intervention. The result is stored with the student record so that changes in risk can be observed over time.

4.5 ANALYTICS DASHBOARD & REPORTING

The Analytics Dashboard & Reporting module presents the results of the system to faculty members and academic administrators. It reads the stored student data, predictions and risk scores and converts them into visual summaries and reports that can be understood at a glance. The dashboard supports the following views:

- **Student performance trends:** progress of assessments and examination results of a student or a group over time.
- **Attendance analytics:** attendance patterns of students and their relationship to risk.
- **Risk distribution:** number and proportion of students at Low, Medium and High risk.
- **At-risk student identification:** a list of students at High and Medium risk, ordered by risk score.

- **Prediction results:** performance and dropout predictions with contributing factors for each student.
- **Visualizations:** charts and tables that make academic trends and risk factors easier to interpret.
- **Faculty dashboard:** a view in which faculty see the students under their responsibility and the students needing attention.
- **Dropout risk analysis:** counts of students in the Stable, Moderate and Critical categories, with filters by risk level and department, and a card for each student showing attendance, GPA (on a 10-point scale), performance and risk level, together with an Intervene action, an option to export the list and an option to run batch analysis.
- **ML models explorer:** a view that compares the four algorithms by accuracy, precision, recall and F1-score, shows the confusion matrix of a selected model and its feature importances, and displays the K-Means student clusters.

The reporting function generates summaries for academic review, covering student performance, risk levels, at-risk students and academic trends. Reports help administrators monitor retention and review the effect of interventions. Access to dashboards and reports is limited to authorized users, because they contain sensitive information about individual students.

The dashboard supports early intervention by identifying who needs attention and why, and the faculty member decides what action to take. By presenting all information consistently from a single source, the module reduces dependence on manual monitoring and encourages data-driven decisions.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSIONS

5.1 IMPLEMENTATION RESULTS

This chapter presents the implementation results of EDU AI as reported by the ML Models Explorer of the system. The results are based on a dataset of 248 students. Four machine learning algorithms were used: Logistic Regression for performance prediction, Random Forest for dropout classification, Naïve Bayes for behavioural risk assessment and K-Means Clustering for student segmentation. The accuracy, precision, recall and F1-score of each model are given in Table 5.1 and compared in Figure 5.1. The dataset name and the train–test split or validation method used are [TO BE PROVIDED].

Table 5.1 Performance of the machine learning models

MODEL	ROLE	ACCURACY	PRECISION	RECALL	F1-SCORE
Logistic Regression	Performance prediction	91.2%	89.7%	90.4%	90.0%
Random Forest	Dropout classification	93.8%	92.1%	94.5%	93.3%
Naïve Bayes	Behavioural risk	87.4%	86.1%	88.2%	87.1%
K-Means Clustering	Student segmentation	88.6%	87.3%	89.1%	88.2%

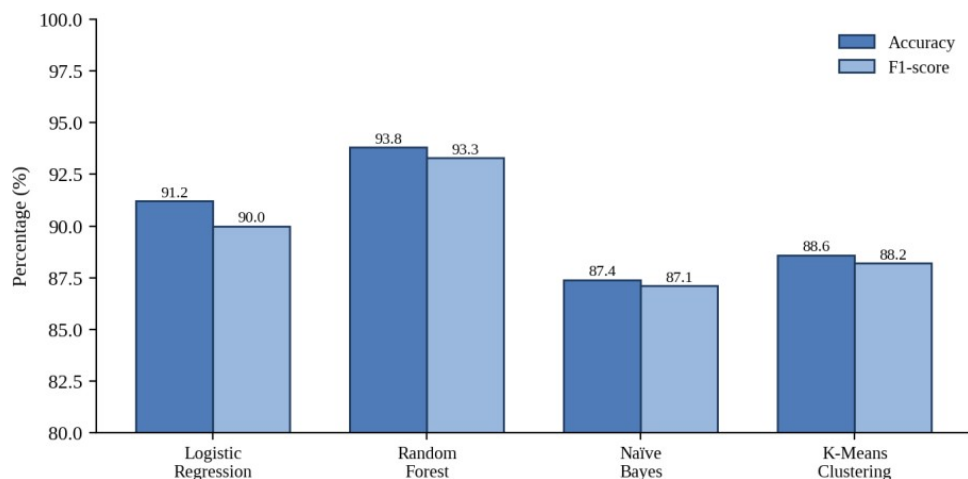


Figure 5.1 Model performance comparison

5.1.1 IMPLEMENTATION OF STUDENT PERFORMANCE PREDICTION

Student performance prediction is implemented using Logistic Regression. The student data was prepared as described in Chapter 4: it was cleaned, missing values were handled, categorical variables were encoded, features were selected or derived and numerical features were scaled where needed. The model forms a weighted linear combination of the features, adds an intercept, passes the result through the sigmoid function and multiplies it by 100 to give a performance score in percent.

The model achieved an accuracy of 91.2%, a precision of 89.7%, a recall of 90.4% and an F1-score of 90.0%. It was the second most accurate of the four models. As each coefficient directly represents the contribution of a feature, the model also helps faculty to see which factors affect a student's predicted performance.

5.1.2 IMPLEMENTATION OF DROPOUT RISK PREDICTION

Dropout risk is predicted by a Random Forest classifier made up of 10 decision trees, each trained on a random subset of features. Each tree votes on the dropout risk class of a student and the majority vote gives the prediction. Random Forest achieved the highest accuracy of the four models, 93.8%, with a precision of 92.1%, a recall of 94.5% and an F1-score of 93.3%.

The confusion matrix of the Random Forest model over the Low, Medium and High risk classes is given in Table 5.2. Of the 207 students whose actual class was Low Risk, 201 were predicted correctly; of the 29 Medium Risk students, 24 were predicted correctly; and of the 12 High Risk students, 10 were predicted correctly.

Table 5.2 Confusion matrix of the Random Forest dropout classifier

ACTUAL CLASS \ PREDICTED CLASS	LOW RISK	MEDIUM RISK	HIGH RISK
Low Risk	201	4	2
Medium Risk	3	24	2
High Risk	1	1	10

Naïve Bayes was used for behavioural risk assessment and achieved an accuracy of 87.4%, a precision of 86.1%, a recall of 88.2% and an F1-score of 87.1%. K-Means Clustering partitioned the 248 students into four clusters for student segmentation; the system reports an accuracy of 88.6%, a precision of 87.3%, a recall of 89.1% and an F1-score of 88.2% for it.

5.1.3 IMPLEMENTATION OF RISK SCORE CLASSIFICATION

The risk score of each student is expressed as a percentage and classified into three levels: Low Risk (below 50%, shown as Stable), Medium Risk (50% to 80%, shown as Moderate) and High Risk (above 80%, shown as Critical). The resulting distribution of the 248 students is presented on the dashboard and summarized in Table 5.3 and Figure 5.2. In total 41 students are flagged for risk, of whom 12 are in the critical category.

Table 5.3 Student risk level distribution

RISK LEVEL	NUMBER OF STUDENTS	PERCENTAGE
LOW RISK (Stable, below 50%)	207	83.5%
MEDIUM RISK (Moderate, 50–80%)	29	11.7%
HIGH RISK (Critical, above 80%)	12	4.8%

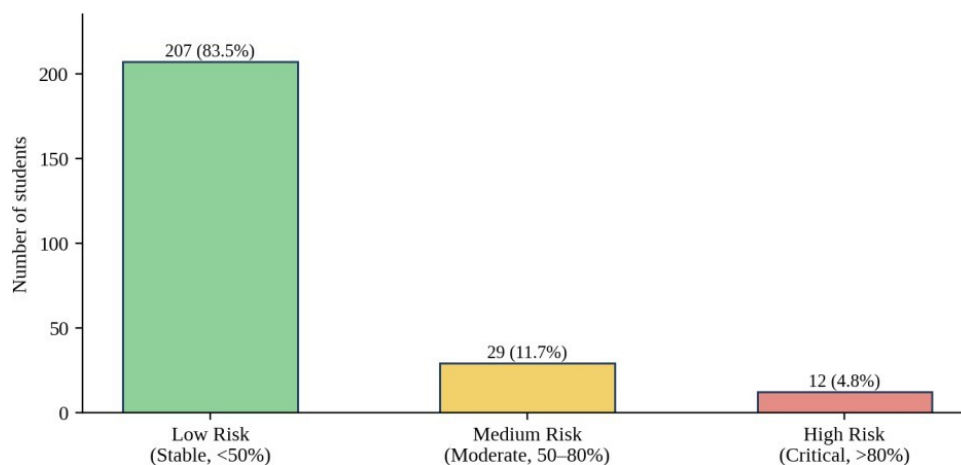


Figure 5.2 Student risk distribution

5.2 DISCUSSION

Random Forest gave the best results on all four measures (accuracy 93.8%, precision 92.1%, recall 94.5% and F1-score 93.3%), which supports its use for dropout classification. Its recall is the highest of the four models, which is important because failing to detect an at-

risk student is usually more costly than monitoring an additional student. Logistic Regression follows with an accuracy of 91.2% and has the advantage that its coefficients are directly interpretable, which suits the performance score. K-Means Clustering (88.6%) and Naïve Bayes (87.4%) were lower in accuracy, and serve supporting purposes, namely student segmentation and behavioural risk assessment. K-Means is an unsupervised method, so the way in which its accuracy, precision, recall and F1-score were computed should be stated: [TO BE PROVIDED].

The confusion matrix shows that most students were placed in their correct class, and that the classes are strongly imbalanced, with 207 Low Risk, 29 Medium Risk and 12 High Risk students. Because the High Risk class is small, a single misclassified student changes its results noticeably. The most serious kind of error is a High Risk student predicted as Low Risk, which occurred once in the matrix, and it is the main reason why recall for the at-risk classes should be examined along with overall accuracy.

The risk levels and contributing factors are intended to support faculty decisions and not to replace them. A High Risk label indicates that attention is advisable and does not imply that a student will certainly fail or leave. The results are based on 248 students; validation on larger and more varied data, and over several academic terms, would help to establish how well they generalize.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

EDU AI is an AI-based student analysis system that helps educational institutions to identify students at risk of poor academic performance or dropout at an early stage. It collects and manages student profiles and academic data, including attendance, assignment scores, quiz scores, examination marks, study hours, previous GPA, demographic information, extracurricular participation and historical dropout records, and prepares them through preprocessing and feature engineering.

The system applies Logistic Regression for performance prediction, Random Forest for dropout classification, Naïve Bayes for behavioural risk assessment and K-Means Clustering for student segmentation, generates an individual risk score for each student, and classifies students into Low, Medium and High risk levels. In the evaluation on 248 students, Random Forest achieved the highest accuracy of 93.8%. An analytics dashboard and reporting function present performance trends, attendance analytics, risk distribution and at-risk students to faculty and academic administrators.

By enabling early identification of at-risk students, EDU AI supports faculty decision making, reduces dependence on manual monitoring, enables timely and targeted intervention and contributes to improved student retention and academic success. The system remains a decision-support tool, in which predictions guide faculty and the final judgement stays with the educator.

6.2 FUTURE SCOPE

The following improvements can extend EDU AI in the future:

- **Larger datasets:** training with data from more students, programmes and academic years to improve reliability and generalization.
- **Real-time student monitoring:** updating predictions as new attendance and assessment data become available.

- **Deep learning models:** evaluating deep learning approaches where sufficient data is available.
- **Explainable AI:** adding stronger explanations of why a student is placed at a given risk level.
- **Integration with college ERP/LMS:** obtaining student data directly from existing institutional systems.
- **Mobile application:** giving authorized faculty access to dashboards on mobile devices.
- **Improved visualization:** adding richer and more interactive charts and report formats.
- **Automated alerts to authorized faculty:** notifying faculty when a student's risk level rises.
- **Continuous model improvement:** periodic retraining, monitoring of model performance and checks for fairness across student groups.
- **Long-term student progress tracking:** following the progress of students across semesters and the outcomes of interventions.

All future development should preserve the privacy of student data, restrict access to authorized users and keep predictions explainable and used as decision support.

REFERENCES

- [1] B. Duro, A. Gomes, F. B. Correia, A. R. Borges, and J. Bernardino, “Machine Learning and Deep Learning for Dropout Prediction in Higher Education: A Review,” *Computers*, vol. 15, no. 3, Art. no. 164, 2026. doi: 10.3390/computers15030164.
- [2] “Predicting Student Dropouts with Machine Learning: An Empirical Study in Finnish Higher Education,” *Technology in Society*, vol. 76, Art. no. 102474, 2024. doi: 10.1016/j.techsoc.2024.102474.
- [3] V. Christou et al., “Performance and Early Drop Prediction for Higher Education Students Using Machine Learning,” *Expert Systems with Applications*, vol. 225, Art. no. 120079, 2023. doi: 10.1016/j.eswa.2023.120079.
- [4] “Multi-Class Phased Prediction of Academic Performance and Dropout in Higher Education,” *Applied Sciences*, vol. 13, no. 8, Art. no. 4702, 2023. doi: 10.3390/app13084702.
- [5] “Statistical and Machine Learning Models for Predicting University Dropout and Scholarship Impact,” *PLOS ONE*, vol. 20, no. 6, Art. no. e0325047, 2025. doi: 10.1371/journal.pone.0325047.
- [6] “A Machine Learning Approach to Predicting Student Dropout and Supporting Timely Intervention,” *Journal of Applied Research in Higher Education*, 2026. doi: 10.1108/JARHE-07-2025-0538.
- [7] “A Real-Time Early Warning Intervention System for At-Risk Students Using Predictive Modeling and Academic Engagement Metrics,” *Procedia Computer Science*, vol. 283, pp. 2486–2495, 2026. doi: 10.1016/j.procs.2026.06.318.
- [8] N. T. Vo, Q. N. Le, and H. Le, “Artificial Intelligence-Driven Early Prediction of Student Dropout and Academic Outcomes in Higher Education: A Comparative Study of Advanced Machine Learning Approaches,” *EAI Endorsed Transactions on Industrial Networks and Intelligent Systems*, 2026. doi: 10.4108/eetinis.131.11758.